227224: metagranodiorite, Edamurta Range

(Yilgarn Craton granites, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Yalgoo (SH 50-2), Mellenbye (2140)
MGA Zone 50, 446991E 6838657N

Sampled on 7 May 2019

This sample was collected from an outcrop southwest of a track on the west side of the Edamurta Range, about 6.3 km east-northeast of Armstrong Well, 5.8 km west of Edamurta Well, and 4.9 km south-southeast of Ederga Well.

Geological context

The unit sampled is a metagranodiorite within the Edamurta greenstone belt, on the northwestern margin of the Yalgoo Dome (Clos et al., 2018; Zibra et al., 2018). A felsic metavolcaniclastic rock, sampled about 4.7 km to the north, yielded a U–Pb zircon date of 2941 ± 3 Ma, interpreted as the as the igneous crystallization age of the dominant zircon-bearing volcanic component (GSWA 207670, Wingate et al., 2019). Metagranitic rocks of the Bunnawarra Suite of the Thundelarra Supersuite in the Yalgoo Dome are similar in age, including 2925 ± 5 Ma tonalite gneiss about 25 km to the east (GSWA 155843, Lu et al., 2020), and 2960 ± 10 Ma migmatitic metatonalite gneiss about 25 km to the southeast (GSWA 207689, Lu et al., 2017).

Petrographic description

The sample is an equigranular, medium-grained metagranodiorite (Fig. 1), consisting of about 50% plagioclase, 30% quartz, 10–12% K-feldspar, 7–8% biotite, <1% magnetite, and trace tourmaline, apatite and zircon. Plagioclase (An_{34}, andesine) is subhedral, equant to tabular, up to 2.5 mm long, and weakly to moderately sericitized. Quartz is anhedral, moderately strained, up to 2.5 mm in diameter, and occurs in part in partially recrystallized, monomineralic aggregates up to 6 mm in size. K-feldspar is anhedral, microperthitic, up to 2 mm long, and shows incipient microcline twinning. Brown–green biotite forms anhedral grains up to 1.5 mm long and also occurs in aggregates of finely recrystallized grains. Some biotite has been replaced by chlorite. Subhedral, mainly equant magnetite grains are up to 0.7 mm in size, and some are partly replaced by hematite or goethite. The thin section contains a single elongate anhedral grain of tourmaline that is 0.6 mm long and shows pink–brown to dark green pleochroism. Euhedral bladed apatite is up to 0.15 mm long. The granodiorite lacks a tectonic foliation but displays evidence of deformation and minor recrystallization at relatively low-grade metamorphic conditions.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mainly subhedral to euhedral. The crystals are up to 300 µm long, and elongate, with aspect ratios up to 9:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, some crystals are dominated by high-U metamict zones, and a few crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 31 July – 1 August 2020, using SHRIMP-B. Eleven analyses of the M257 standard were obtained during the session, of which nine analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.19% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their U and Th concentrations.

Group I comprises 17 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2941 ± 3 Ma (MSWD = 0.86). These analyses indicate mainly moderate concentrations of U (94–385 ppm, median 196 ppm) and Th (37–233 ppm, median 119 ppm).

Figure 1. Hand specimen image of sample 227224: metagranodiorite, Edamurta Range

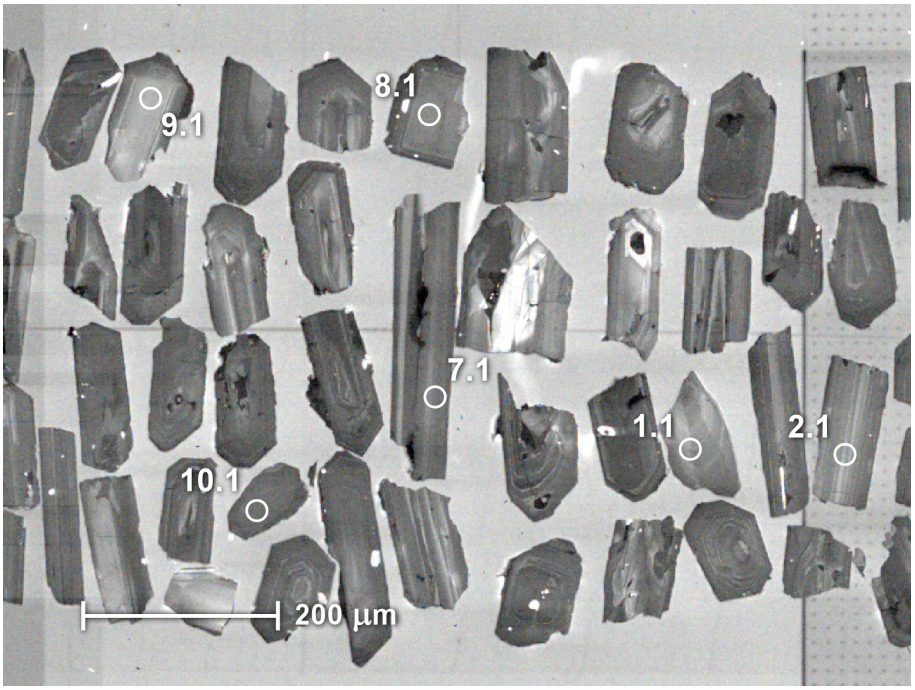

Figure 2. Cathodoluminescence image of representative zircons from sample 227224: metagranodiorite, Edamurta Range. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 227224: metagranodiorite, Edamurta Range

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	16	16.1	163	100	0.62	-0.033	1.749	0.018	0.21278	0.00101	1.748	0.018	0.21307	0.00102	2916	24	2929	8	0.4
I	11	11.1	215	149	0.69	0.031	1.745	0.016	0.21376	0.00086	1.746	0.016	0.21348	0.00087	2919	22	2932	7	0.4
I	5	5.1	96	37	0.39	0.146	1.775	0.022	0.21482	0.00131	1.777	0.022	0.21352	0.00139	2878	29	2932	11	1.9
I	3	3.1	185	133	0.72	0.027	1.752	0.017	0.21391	0.00150	1.753	0.017	0.21368	0.00151	2910	23	2934	11	0.8
I	1	1.1	134	49	0.37	0.000	1.770	0.019	0.21389	0.00110	1.770	0.019	0.21389	0.00110	2888	25	2935	8	1.6
I	17	17.1	335	167	0.50	0.149	1.890	0.016	0.21541	0.00075	1.893	0.016	0.21408	0.00079	2734	19	2937	6	6.9
I	4	4.1	217	90	0.41	0.097	1.784	0.016	0.21495	0.00084	1.786	0.016	0.21409	0.00088	2866	21	2937	7	2.4
I	6	6.1	136	57	0.42	0.024	1.779	0.019	0.21487	0.00105	1.779	0.019	0.21466	0.00106	2876	25	2941	8	2.2
I	14	14.1	216	83	0.38	0.008	1.775	0.017	0.21476	0.00088	1.775	0.017	0.21469	0.00088	2880	22	2941	7	2.1
I	12	12.1	318	189	0.59	0.000	1.772	0.015	0.21471	0.00075	1.772	0.015	0.21471	0.00075	2885	20	2941	6	1.9
I	8	8.1	196	119	0.61	-0.034	1.762	0.017	0.21443	0.00090	1.762	0.017	0.21473	0.00091	2898	23	2942	7	1.5
I	18	18.1	119	153	1.29	0.288	1.904	0.022	0.21737	0.00120	1.910	0.022	0.21481	0.00134	2715	26	2942	10	7.7
I	7	7.1	260	132	0.51	0.000	1.764	0.016	0.21495	0.00080	1.764	0.016	0.21495	0.00080	2895	21	2943	6	1.6
I	15	15.1	334	233	0.70	0.092	1.788	0.015	0.21619	0.00075	1.790	0.015	0.21537	0.00077	2861	20	2946	6	2.9
I	9	9.1	94	57	0.61	-0.020	1.785	0.022	0.21542	0.00136	1.784	0.022	0.21559	0.00137	2868	29	2948	10	2.7
I	10	10.1	385	207	0.54	0.069	1.754	0.014	0.21669	0.00070	1.756	0.014	0.21607	0.00072	2906	19	2952	5	1.5
I	2	2.1	124	78	0.63	0.000	1.797	0.020	0.21610	0.00111	1.797	0.020	0.21610	0.00111	2852	25	2952	8	3.4
P	13	13.1	734	411	0.56	0.041	1.729	0.012	0.21288	0.00048	1.730	0.012	0.21252	0.00048	2941	17	2925	4	-0.6

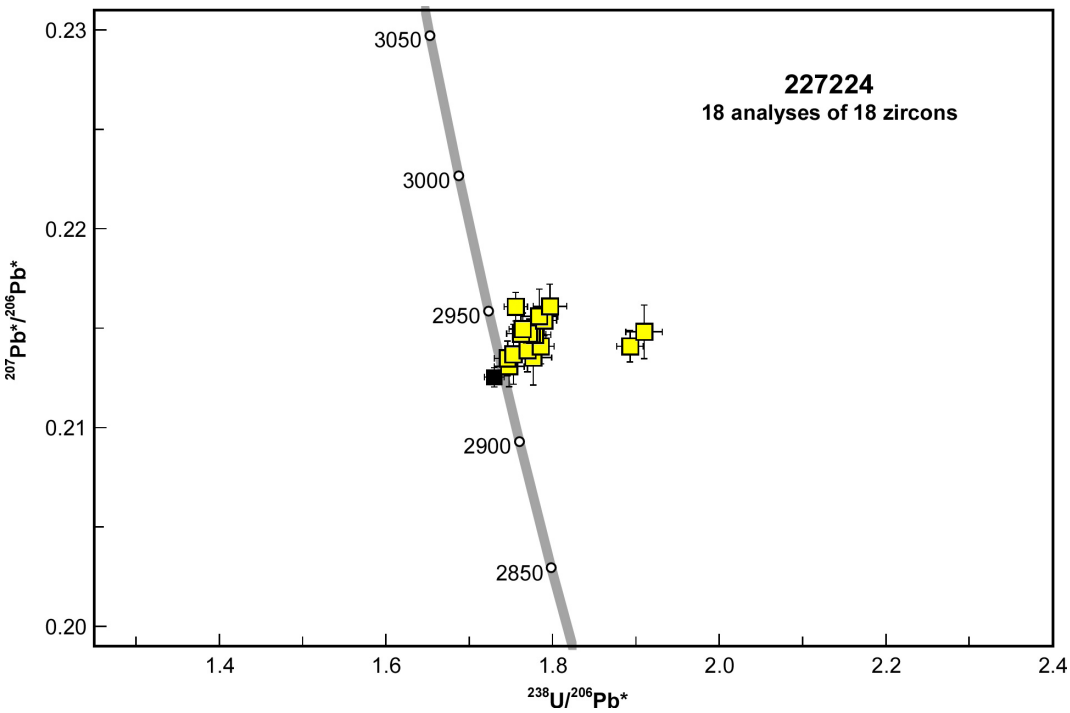

Figure 3. U–Pb analytical data for sample 227224: metagranodiorite, Edamurta Range. Yellow squares indicate Group I (magmatic zircons); black square indicates Group P (radiogenic-Pb loss)

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2925 ± 4 Ma (1σ). This analysis indicates high concentrations of U (734 ppm) and Th (411 ppm).

The date of 2941 ± 3 Ma for the 15 analyses in Group I is interpreted as the magmatic crystallization age of the tonalite. The date of 2925 ± 4 Ma (1σ) for the single analysis in Group P is interpreted to reflect minor ancient loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, TJ 2021, 227224: tonalite, Edamurta Range; Geochronology Record 1803: Geological Survey of Western Australia, 4p.

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